

Exercises 09/16

1. Consider the following IS-LM Model

$$Y = C(Y - T, r) + I(r, Y) + G$$

$$\frac{M}{P} = L(Y, r)$$

The endogenous variables are Y and r , and the exogenous variables are T , G , M , and P . Assume $0 < \text{MPC} + I_Y < 1$, $C_r < 0$, $I_r < 0$, $L_r < 0$, $L_Y > 0$. Use the implicit function theorem to compute the signs of $\frac{\partial r}{\partial M}$ and $\frac{\partial Y}{\partial T}$.

2. Consider the following consumer choice problem:

$$\begin{aligned} \max \quad & u(x, y) \\ & p_x x + p_y y = I \end{aligned}$$

where $u(x, y) = x + h(y)$, with $h' > 0$ and $h'' < 0$. Apply the implicit function theorem to the first order conditions to compute $\frac{\partial y}{\partial p_x}$.

3. Consider the following consumer choice problem:

$$\begin{aligned} \max \quad & u(c_1) + u(c_2) \\ & c_1 + s = y_1 \\ & c_2 = (1 + r)s + y_2 \end{aligned}$$

where $u(c) = \frac{1}{1-\sigma}(c^{1-\sigma} - 1)$, and $\sigma > 0$. Here, think of c_1 as consumption in period 1, and c_2 as consumption in period 2; also, y_i is income in period i and r is the interest rate on savings s . Use the implicit function theorem to determine the range of values for σ for which $\frac{\partial c_1}{\partial r} > 0$.

4. Show that finite set of orthogonal vectors is linearly independent.
5. Show that if $\mathbf{v}, \mathbf{w} \in \mathbf{V}$ then there exists $\alpha \in \mathbb{R}$ so that $\mathbf{v} - \alpha\mathbf{w}$ is orthogonal to $\mathbf{w}$.